M378K: April 24th, 2026.

Sufficient statistics.

Consider two election candidates: A & B

Goal: To "predict" if A win.

Let the random sample be $Y_1, Y_2, \dots, Y_n$

Scenarios for book-keeping:

- $T_1 = (Y_1, \dots, Y_n)$
- $T_2 = Y_1 + \cdots + Y_n$
- $T_3 = \mathbb{1}_{\{\bar{Y} > 0.5\}}$

Formally,
- the conditional dist'n of $T_1 = (Y_1, \dots, Y_n)$ given $T_2 = Y_1 + \cdots + Y_n$ does not depend on p
- the conditional dist'n of $T_1 = (Y_1, \dots, Y_n)$ given $T_3 = \mathbb{1}_{\{\bar{Y} > 0.5\}}$ DOES DEPEND ON p.

DISCRETE

In general, consider two r.v.s Y and T, the conditional dist'n of Y given T are these probabilities $\mathbb{P}[Y = y \mid T = t]$

In the discrete case, take $(Y_1, \dots, Y_n)$

$$p_{Y_1, \dots, Y_n, T}(y_1, \dots, y_n, t) = \mathbb{P}[Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n, T = t]$$

Def'n. The conditional joint pmf for t such that $\mathbb{P}[T=t]>0$ is

$$p_{Y_1,\dots,Y_n|T}(y_1,\dots,y_n|t) = \mathbb{P}[Y_1=y_1,\dots,Y_n=y_n|T=t]$$
$$= \frac{\mathbb{P}[Y_1=y_1,\dots,Y_n=y_n,T=t]}{\mathbb{P}[T=t]}$$

Analogously, in the continuous case

$$f_{Y_1,\dots,Y_n|T}(y_1,\dots,y_n|t) = \frac{f_{Y_1,\dots,Y_n,T}(y_1,y_2,\dots,y_n,t)}{f_T(t)}$$

Def'n. A statistic T of a random sample $(Y_1,\dots,Y_n)$ is said to be SUFFICIENT for an unknown parameter θ if the conditional dist'n of the random sample $(Y_1,\dots,Y_n)$ given T does not depend on θ.

Example. $\boxed{\text{Claim:}}$ $T_2 = Y_1+\dots+Y_n$ is sufficient for p

$$p_{Y_1,\dots,Y_n|T_2}(y_1,\dots,y_n|t) = \frac{\mathbb{P}[Y_1=y_1,\dots,Y_n=y_n,T_2=t]\checkmark}{\mathbb{P}[T_2=t]\checkmark}$$

Cases:

1st $y_1+\dots+y_n \neq t \longrightarrow$ we get 0

2nd $\boxed{y_1+\dots+y_n=t}$ $\{T_2=t\} \supseteq \{Y_1=y_1,\dots,Y_n=y_n\}$

$$\checkmark\ \mathbb{P}[Y_1=y_1,\dots,Y_n=y_n,\boxed{T_2=t}] = \mathbb{P}[Y_1=y_1,\dots,Y_n=y_n]$$
$$= \mathbb{P}[Y_1=y_1]\cdots\mathbb{P}[Y_n=y_n]$$
$$= p^{y_1}(1-p)^{1-y_1}\,p^{y_2}(1-p)^{1-y_2}\cdots p^{y_n}\cdot(1-p)^{1-y_n}$$

$$= p^{\sum y_i}(1-p)^{n-\sum y_i}$$

$$= p^t (1-p)^{n-t}$$

$$\checkmark\ \mathbb{P}[T_2 = t] = \binom{n}{t} p^t (1-p)^{n-t}$$

$$p_{Y_1,\dots,Y_n \mid T_2}(y_1,\dots,y_n \mid t) = \frac{\cancel{p^t (1-p)^{n-t}}}{\binom{n}{t}\cancel{p^t (1-p)^{n-t}}} = \frac{1}{\binom{n}{t}}$$

Theorem. $\boxed{\text{The Fisher-Neyman Factorization Criterion}}$

Let $Y_1,\dots,Y_n$ be a random sample w/ the likelihood f'tion

$$L(\theta; y_1,\dots,y_n).$$

A statistic T is sufficient for θ if and only if L can be expressed as

$$L(\theta; y_1,\dots,y_n) = g(\theta, T(y_1,\dots,y_n)) \cdot h(y_1,\dots,y_n)$$

Example. $\boxed{\text{Bernoulli}}$ $\qquad \theta \leftrightarrow p$

$$L(p; y_1,\dots,y_n) = p^{y_1}(1-p)^{1-y_1} p^{y_2}(1-p)^{1-y_2} \cdots p^{y_n}(1-p)^{y_n}$$

$$= p^{\sum y_i = t}(1-p)^{n-\sum y_i = t}$$

$$= p^t (1-p)^{n-t}$$

$$g(p,t) = p^t(1-p)^{n-t} \text{ and } h \equiv 1$$